

Executive Summary

Nova Desk is a custom, production-grade SaaS application designed for enterprise hardware procurement, software licensing, and cloud permission workflows. Engineered to full commercial POC requirements with hierarchical Role-Based Access Control (RBAC), multi-step approval stages, real-time audit logging, responsive dark-mode styling with Lenis smooth scrolling, strict zero-credential-leakage security, and self-contained Docker orchestration.

CORE DEPLOYMENT & REPOSITORY ACCESS

LIVE PRODUCTION ENVIRONMENT

Live URL: <https://saas.radhesh.me>
Status: Active / Deployed
SSL / Edge: Secure TLS Termination
Rendering: Next.js 15 Standalone Optimized

SOURCE REPOSITORY

GitHub Remote: github.com/radheshpai87/SaaSRBAC
Default Branch: master
Build Validation: 11/11 Routes Compiled (0 Errors)
Automated Tests: 15/15 Passing

ARCHITECTURAL HIGHLIGHTS & ROLE-BASED ACCESS (RBAC)

ADMIN OPERATIONS CONSOLE

- Full inventory oversight & user provisioning
- Departmental spending ceiling controls
- System-wide immutable audit trail inspection
- Status toggle & role privilege management

MANAGER APPROVAL QUEUE

- Departmental request review & triage
- One-click budget approval & fiscal sign-off
- Rejection workflows with feedback notes
- Real-time team expenditure monitoring

USER PORTAL

- Hardware, software, & cloud access requests
- Vendor cost calculations & ROI payback terms
- Status tracking with priority indicators
- Personal spending limit visibility

TECHNICAL STACK & INFRASTRUCTURE

Layer	Technology	Implementation Details
Framework	Next.js 15 (App Router) + React 19	Server Actions, Streaming SSR, dynamic OpenGraph image generator
Security & Auth	Auth.js (NextAuth v5) + Edge Middleware	Encrypted JWT sessions, Bcrypt hashing, strict role boundary protection
Database & ORM	PostgreSQL + Prisma ORM + PGlite	Zero-friction embedded database engine with external PostgreSQL driver support
UI & Experience	Tailwind CSS + Lenis + Lucide Icons	Linear-grade dark palette, smooth momentum scrolling, 100% SVG icons (0 emojis)
Containerization	Docker + Docker Compose	Multi-stage Alpine image, automated migration & seed startup entryptoint

KEY ENGINEERING DELIVERABLES COMPLETED TODAY

SECURITY & CREDENTIAL SANITIZATION

- 100% credential sanitization across code bundles, UI states, and git history.
- Secure server-side password hashing using bcryptjs with salt rounds.
- Demo account switcher pre-fills credentials without exposing keys in payloads.

DESIGN & UX REFINEMENTS

- Completely eradicated all emojis in favor of crisp Lucide React SVG icons.
- Implemented Lenis Smooth Scrolling for fluid, high-end momentum physics.
- Streamlined navigation, removed top clutter, and unified into a single-line footer.

DYNAMIC OPENGGRAPH & METADATA

- Built dynamic server-side OpenGraph generator at /opengraph-image.
- Automated Twitter/X summary large card generation at /twitter-image.
- Social metadata preview cards with branding and role highlights.

ZERO-CONFIG SELF-CONTAINED DOCKER

- Multi-stage Dockerfile creating a hardened, non-root Alpine container.
- Automated container entrypoint running schema migrations & seeds on startup.
- Instant home server launch with: docker compose up --build -d.

GIT REPOSITORY COMMIT HISTORY

Commit Hash	Message	Key Deliverable
40c0262	feat(docker): add automated container entrypoint and self-contained zero-config startup	Added docker-entrypoint.sh, auto database migration & seed on fresh start
9d07e9a	style/footer): convert footer to single clean minimalist line and align navbar links	Single line footer layout, removed redundant links, streamlined navigation
54d59ee	refactor(ui): remove top banner and versioning tags	Purged version tags and top operational banners for clean client aesthetic
b932be0	refactor(security): sanitize all passwords, remove emojis, enhance nav UI	Zero emoji enforcement, sanitized credentials, enhanced dark navigation
a29db00	feat(seo): add dynamic OpenGraph and Twitter card image generation with social metadata	Generated /opengraph-image and /twitter-image endpoints with custom Canvas
8563d92	feat(docker): add production multi-stage Dockerfile and docker-compose deployment	Multi-stage Alpine runner, healthchecks, persistent data volumes
03076ca	feat(ui): add Lenis smooth scrolling, modern dark aesthetic with glow visuals	Integrated Lenis smooth scrolling engine and Linear/Vercel visual styling

DEPLOYMENT & VERIFICATION INSTRUCTIONS

```
# 1. Clone Repository
git clone https://github.com/radheshpai87/SaaS RBAC.git && cd SaaS RBAC

# 2. Launch Container (Automated Zero-Config Startup)
docker compose up --build -d

# 3. Access Live System
Local: http://localhost:3000 | Production: https://saas.radhesh.me
```